Mohamed Rissal Hedna

AI Researcher | LLM Calibration, Uncertainty Quantification & Trustworthy AI

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SUMMARY

Researcher working on uncertainty and calibration in large language models. First author of SECL (EMNLP 2026 Main Conference), a test-time training approach to LLM confidence calibration. Interested in whether models can know when they are wrong and whether they can reason causally. Alongside research, two years of industry work on hallucination detection and evaluation for agentic LLM systems in regulated life sciences.

Research interests: uncertainty quantification and calibration, probabilistic ML, trustworthy LLMs and agents, hallucination detection, causality in language models

EDUCATION

M.Sc. Intelligent Adaptive Systems, University of Hamburg — Grade 1.49 (1.0 best) | Top 1% of cohort | 2026

Thesis: SECL: Self-Calibrating Language Models (advisors: Prof. Chris Biemann, Jan Strich) | Focus: neural networks, knowledge processing, statistical NLP

B.Sc. Computer Science, Lebanese American University — CGPA 3.91/4.0 | Top 1% of cohort | High Distinction

Minor in Data Analytics | MEPI Tomorrow's Leaders scholarship | 2019 – 2023

Relevant coursework: probability and statistics, linear algebra, calculus I–III, discrete structures, machine learning, data mining, neural networks, statistical methods of language technology, knowledge processing

PUBLICATIONS

M. R. Hedna, J. Strich, M. Semmann, C. Biemann. "SECL: Self-Calibrating Language Models via Test-Time Discriminative Distillation." **EMNLP 2026 (Main Conference)**, Budapest. arXiv:2604.09624 | code | project page

M. R. Hedna, S. S. Nder. "Adversarial Patch Attacks on Vision-Based Cargo Occupancy Estimation via Differentiable 3D Simulation." Preprint, arXiv:2511.19254, 2025.

M. R. Hedna, Y. Djemmal, K. Mershad. "A Model for the Automatic Mixing of Multiple Audio and Video Clips." IEEE International Conference on Cyberworlds, 2023. code

RESEARCH EXPERIENCE

University of Hamburg, Language Technology Group — Hamburg | 2025 – 2026

Graduate Researcher (advisors: Prof. Chris Biemann, Jan Strich)

- Developed SECL, a test-time training method for confidence calibration that uses the model's own generation–discrimination gap as label-free self-supervision through an online LoRA adapter (PyTorch, PEFT; single A100 80 GB and RTX A6000, 2 to 4 GPU-hours per run).
- Reduced Expected Calibration Error by 56 to 78% across four LLMs (three model families) and four domains while training on only 6 to 26% of the question stream, validated with seven component ablations.
- Built a differentiable 3D adversarial-patch pipeline in Mitsuba 3 against a cargo-occupancy classifier, reaching 84.9% attack success versus a weaker 2D compositing baseline.
- Wrote a survey on causality in language models as part of the Knowledge Processing graduate module.

Advanced Telerobotics Lab, Kent State University — Ohio, USA | 2022

Visiting Research Assistant

- Built a custom C/Python CNN optimized for GPU inference on embedded NVIDIA Jetson, reaching sub-50ms latency for obstacle detection in GPS-denied environments, with LiDAR-vision fusion under tight memory and power budgets.

INDUSTRY EXPERIENCE

PHAROS Labs — Hamburg | 2024 – Present

AI Engineer, Forward Deployed (Aug 2026 – Present) | AI Engineer, Part-Time (2024 – Jul 2026)

- Designed and deployed hallucination-detection layers for agentic medical-writing features in regulated life sciences, where every claim must be verified by domain experts, reducing unsupported answers by ~78% on production evaluation data.
- Built automated evaluation and monitoring for reasoning- and retrieval-heavy RAG pipelines, and benchmarked retrieval backends on latency, recall and cost before migrating production retrieval to pgvector.

Fiindo (Contract, 2025) | AdaLab UG (ML Developer, 2023 – 2024) — Hamburg

- Built an agentic pipeline turning raw financial data into narrated short-form video; raised peak-load throughput ~60% on a multimodal generation platform serving 500+ active users.

TEACHING, TALKS & SERVICE

Teaching: Calculus peer mentor, Lebanese American University | Online instructor teaching Python and machine learning to students | Python from A to Z (18-video YouTube course in Algerian Arabic)

Talks: SECL poster presentation at EMNLP 2026 (Budapest) | Speaker at code.talks (Hamburg, 2026) and LLMday | **Service:** student volunteer, EMNLP 2026

Community: Founder of withrssle (2021 – present), an education platform helping North African students access scholarships, with a 130K audience; organized an AI tour across Algeria

SKILLS

Tools: Python, PyTorch, Hugging Face (Transformers, PEFT), C, SQL | **Languages:** English (C2), French (C1), German (B2), Arabic (native)